# Homework 09

The kernel-tuning question moved here from Homework 08. A solution for Question 1 will be available after Monday, October 26, 2026. No solution will be provided for Question 2.

## Question 1: Choosing an SVM kernel

Use `saheart.csv` from the combined ZIP. As in Homework 7, predict `chd` using the same nine predictors, coding `famhist` as 1 for `Absent` and 2 for `Present`.

Compare SVMs with a linear kernel, polynomial kernels of degrees 2 (quadratic), 3, and 4, and a radial basis kernel. Choose **no more than 20 candidate settings in total** across these kernels. Vary the cost $C$ for each kernel and the radial scale $\gamma$ for the radial kernel. State any other kernel settings you hold fixed.

Divide the observations into five folds yourself and use the same folds for all candidates. Standardize predictors using only the fitting observations in each fold. Report the candidate settings and their mean cross-validation misclassification rates. Which kernel and settings do you prefer? Give a brief justification.

### Answer

| Kernel | Degree | C | Gamma | CV_Error | SE |
|---|---|---|---|---|---|
| linear | NaN | 0.1 | None | 0.266199 | 0.006963 |
| linear | NaN | 1.0 | None | 0.272744 | 0.007263 |
| linear | NaN | 10.0 | None | 0.277045 | 0.004070 |
| poly | 2.0 | 0.1 | scale | 0.270547 | 0.018932 |
| poly | 2.0 | 1.0 | scale | 0.270477 | 0.015637 |
| poly | 2.0 | 10.0 | scale | 0.294273 | 0.012800 |
| poly | 3.0 | 0.1 | scale | 0.274801 | 0.018820 |
| poly | 3.0 | 1.0 | scale | 0.311547 | 0.023245 |
| poly | 3.0 | 10.0 | scale | 0.346213 | 0.015787 |
| poly | 4.0 | 0.1 | scale | 0.294320 | 0.016918 |
| poly | 4.0 | 1.0 | scale | 0.322394 | 0.017027 |
| poly | 4.0 | 10.0 | scale | 0.354862 | 0.015098 |
| rbf | NaN | 0.1 | 0.01 | 0.346237 | 0.023671 |
| rbf | NaN | 0.1 | 0.1 | 0.339715 | 0.029685 |
| rbf | NaN | 1.0 | 0.01 | 0.277022 | 0.015373 |
| rbf | NaN | 1.0 | 0.1 | 0.283497 | 0.015533 |
| rbf | NaN | 10.0 | 0.01 | 0.266199 | 0.016740 |
| rbf | NaN | 10.0 | 0.1 | 0.315942 | 0.022989 |
| rbf | NaN | 100.0 | 0.01 | 0.279149 | 0.016949 |
| rbf | NaN | 100.0 | 0.1 | 0.348387 | 0.017406 |

```
Best model:
Kernel        linear
Degree           NaN
C                0.1
Gamma           None
CV_Error    0.266199
SE          0.006963
Name: 0, dtype: object
```

### Model Selection and Justification

I compared 20 SVM candidates using five-fold cross-validation with the same folds for all models (seed 2026). Predictors were standardized using only the training observations in each fold. For polynomial kernels, I fixed `gamma="scale"` and `coef0=1`, while varying the degree and cost $C$. All other parameters were kept at their scikit-learn defaults.

The **linear SVM with** $C = 0.1$ and **RBF SVM with** $C = 10, \gamma = 0.01$ tied for the lowest mean CV misclassification rate of **0.2662**. I prefer the linear SVM because it is simpler and has a smaller standard error across folds (0.0070 versus 0.0167), suggesting more consistent performance. Higher-degree polynomial kernels generally performed worse, particularly at larger values of $C$. Thus, the additional complexity of nonlinear kernels did not provide a clear improvement in predictive performance.

## Question 2: Write and use a short SVM analysis skill

In Homework 8, you reviewed another student's skill. Review the comments and suggestions you received on your own Homework 7 skill. Use what you learned from both sides of the peer review to write a new, shorter skill for SVM model fitting.

**Write the skill by hand.** Create a `SKILL.md` with a short name and description in YAML frontmatter. Write the entire file yourself, without asking AI to draft or edit it. The instruction body after the closing `---` of the frontmatter must be **no more than 50 lines**, including blank lines. Keep the skill independent of any particular dataset, variable names, or observed results, and load it only when you explicitly invoke it.

Your skill should direct the agent to:

1. judge whether the data are suitable for an SVM analysis;
2. choose which kernels to consider and explain why;
3. identify tuning parameters and a method for tuning them;
4. choose an evaluation method that supports selecting a final model;
5. require a compact figure and table that justify the choices;
6. interpret the fit and report the key numerical results; and
7. prioritize the evidence needed to justify the final model in **no more than one printed page at 11-point font**.

**Apply the skill.** Explicitly invoke your skill to analyze `saheart.csv`, using the same response, nine predictors, and `famhist` coding as in Question 1. Provide these data details when invoking the skill, not in the skill itself. Let the skill guide the analysis, then check the resulting report and calculations.

**Submit the results.** Include the one-page, 11-point analysis report in your Homework 09 submission and append the complete contents of your `SKILL.md` afterward. The appended skill is outside the one-page limit.

### Answer

The skill was invoked as `/svm-analysis` with the data details (response `chd`, the nine predictors, and the `famhist` coding) supplied in the prompt. The one-page, 11-point report it produced follows; the analysis code is `q2_svm_analysis.py`.

#### Skill Development and Verification

**Peer-review feedback and revisions**

Based on feedback received on my Homework 7 skill and my review of another student's skill, I improved the reproducibility and consistency of the SVM analysis instructions. After the initial run, I revised the skill to specify a fixed 17-candidate grid, stratified five-fold cross-validation with seed 2026, and an explicit one-standard-error selection rule. I also required class-specific evaluation and reproducible code.

**Report verification**

I checked the cross-validation results and out-of-fold predictions. The quadratic SVM with $C = 0.1$ achieved the lowest mean CV misclassification rate of 0.2662 (SE = 0.0123). Seven candidates were within one standard error of the minimum.

The confusion matrix contained TN = 273, FP = 29, FN = 94, and TP = 66, giving CHD recall of 41.25% and non-CHD recall of 90.40%. These results confirm the report's conclusions and highlight the model's limited sensitivity to CHD cases.

**Difference from Question 1**

Questions 1 and 2 selected different models because they used different cross-validation fold assignments and candidate grids. Question 1 selected a linear SVM with $C = 0.1$, while Question 2 selected a quadratic SVM with $C = 0.1$. Although Question 2 used a one-standard-error selection rule, the quadratic model also had the lowest mean CV error, so the rule did not change the final selection.

## SVM Analysis of CHD Risk (saheart.csv)

**Data.** 462 subjects, 9 predictors (famhist coded Absent = 1, Present = 2), binary response chd (160 cases, 34.6%; 0 missing values). All predictors are numeric after coding, so an SVM is applicable; the classes are moderately imbalanced and the largest predictor correlation is adiposity–obesity (0.72). Scaling is fit within each training fold.

**Kernels.** Linear: simplest baseline; RBF: flexible baseline for general nonlinearity; quadratic: plausible pairwise risk-factor interactions. Degrees 3–4 excluded: no reason to expect higher-order interactions at n = 462.

**Design.** Default grid, 17 candidates, on identical stratified 5-fold splits (shuffle, seed 2026): linear (C = 0.01–10), RBF (C = 1, 10, 100 × $\gamma$ = 0.003, 0.01, 0.03), quadratic polynomial (C = 0.01–10; fixed $\gamma$ = "scale", coef0 = 1). No deviations from the defaults; other SVC settings at their defaults. **Rule:** among models within one SE of the lowest CV error, choose the simplest kernel (linear < poly < RBF), then the smallest C and $\gamma$.

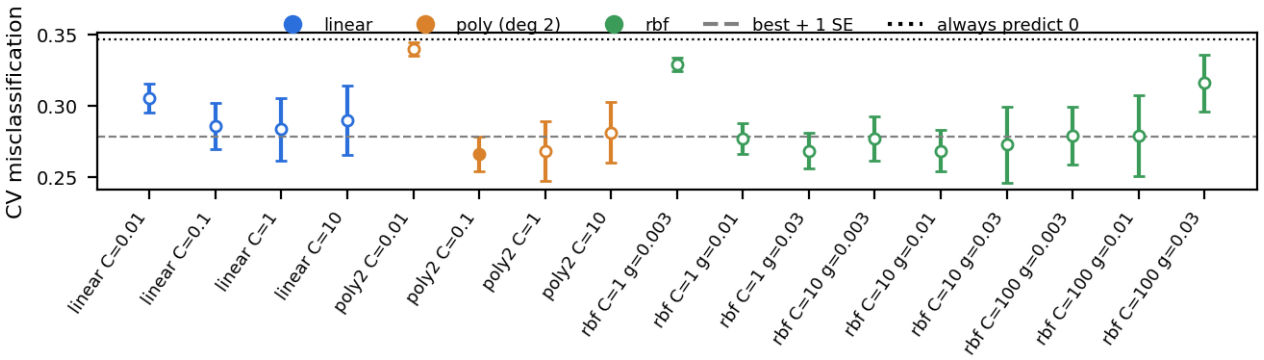



**Table: five best candidates (out-of-fold).**

| Model | CV error | SE | Recall chd=1 | Recall chd=0 |
|---|---|---|---|---|
| poly2 C=0.1 | 0.266 | 0.012 | 0.412 | 0.904 |
| poly2 C=1 | 0.268 | 0.021 | 0.469 | 0.871 |
| rbf C=1 g=0.03 | 0.268 | 0.012 | 0.444 | 0.884 |
| rbf C=10 g=0.01 | 0.268 | 0.015 | 0.481 | 0.864 |
| rbf C=10 g=0.03 | 0.273 | 0.026 | 0.475 | 0.861 |

**Selected model: poly2 C=0.1** (quadratic polynomial, coef0 = 1). It has the lowest CV error, 0.266 (SE 0.012). 7 candidates fall within one SE ($\leq$ 0.278), all of them polynomial or RBF, so the nonlinear models cannot be told apart. No linear model meets the threshold (best linear: 0.283). Among the tied models, the quadratic kernel is the simplest and has one of the smallest fold-to-fold SEs.

**Class-specific results (out-of-fold).** TN = 273, FP = 29, FN = 94, TP = 66. Recall is 90.4% for non-cases but only 41.2% for CHD cases: 94 of 160 cases are missed. The 0.266 overall error therefore overstates how well cases are detected (always predicting 0 gives 0.346).

**Limitations.** The top seven models are within one SE, so the choice is not decisive. Tuning targeted misclassification, not case detection; class weights or a lower threshold would raise case recall at the cost of more false positives. With no outer test set, the CV error of the selected model is slightly optimistic.

*Reproducibility:* Python 3.13.12, scikit-learn 1.8.0, numpy 2.4.3, pandas 3.0.1; code in `q2_svm_analysis.py`.

**Appendix: complete contents of `SKILL.md`**

```
---
name: svm-analysis
description: >
  Guides reproducible SVM classification, kernel selection,
  hyperparameter tuning, evaluation, and concise reporting.
  Use only when explicitly invoked.
disable-model-invocation: true
---

# SVM Classification Analysis

## 1. Data Assessment
- Inspect sample size, predictors, missing values, class balance, and variable types.
- Confirm SVM classification is appropriate; explain limitations.
- Encode categorical predictors and standardize without data leakage.

## 2. Kernel Selection
- Always include linear and RBF kernels.
- Include degree-2 polynomial kernels to assess plausible pairwise interactions.
- Consider higher degrees only with justification and a revised grid.
- Explain why each kernel is included or excluded.
- Record all candidates and fixed kernel parameters.

## 3. Tuning and Validation
- Default grid:
  - Linear: C = [0.01, 0.1, 1, 10].
  - RBF: C = [1, 10, 100],
    gamma = [0.003, 0.01, 0.03].
  - Degree-2 polynomial: C = [0.01, 0.1, 1, 10],
    gamma="scale", coef0=1.
- Use stratified 5-fold CV, shuffle=True, seed=2026.
- Use identical folds for all candidates.
- Fit preprocessing only on training folds.
- Record any deviations from these defaults.

## 4. Model Selection
- Calculate mean CV misclassification and SE across folds.
- Retain candidates with CV error <= best error + best SE.
- Select the simplest eligible kernel:
  linear, degree-2 polynomial, then RBF.
- Within a kernel, select the smallest C, then gamma.
- Report the selected model's out-of-fold
  confusion matrix and class-specific recall.
- Discuss limitations of minority-class detection.

## 5. Reporting
- Include a compact table of leading models, tuning settings, CV errors, and variability.
- Include a figure comparing candidate performance.
- Report the selected kernel, hyperparameters, error, and class-specific results.
- Explain why the selected model is preferred and identify limitations.
- Record software versions and provide reproducible code separately.
- Prioritize evidence supporting model selection.
- Keep the main report within one printed page at 11-point font.
```